Preserving Visual Evidence in Compressed Multimodal Reasoning

Research Proposal

Abstract

Compressing multimodal reasoning can reduce output length while leaving unclear whether a short explanation preserves the visual evidence supporting its answer. We propose image-conditioned sufficiency supervision for concise traces that explicitly retain necessary premises and valid deductions. A bounded study will use one 7B answering model, annotated diagram problems, and human-checked image pairs. A separate judge will score explanatory support rather than infer sufficiency from answer correctness. Controlled experiments will compare evidence and style rewards on identical rewritten data, with matched training compute and achieved reasoning-token budgets. Original-trace and length-preserving rewrite controls will probe alternative explanations. We hypothesize improved premise retention and accuracy under compression; the study will also identify budgets where the proposed supervision offers no advantage.

1 Introduction

Chain-of-thought prompting makes intermediate reasoning explicit, while reinforcement learning can improve reasoning performance [1, 2]. In visual mathematics, a short solution must retain facts read from diagrams alongside algebraic deductions. Removing "the marked angle is 40°" can leave an otherwise plausible calculation unsupported. Whether compression preferentially removes these visual premises remains an empirical question.

SketchThinker-R1 [3] learns concise traces using a style judge that sees neither the image nor the question. Consequently, its style supervision does not establish preservation of answer-critical visual evidence. Its Table 2 reports a MathVision decrease at 3B, from 26.9 to 25.3 accuracy points. This reported comparison motivates testing accuracy boundaries, but does not establish a general compression failure.

The central question is whether evidence-aware supervision improves the accuracy–length tradeoff beyond style supervision. Three research questions organize the proposal: (RQ1) are necessary visual premises deleted more often than necessary nonvisual steps? (RQ2) does sufficiency supervision improve accuracy and premise retention at matched average reasoning-token budgets, including harder items? (RQ3) does any advantage persist when rewritten data and training compute are controlled? Planned contributions are an operational sufficiency criterion, an annotated intervention set with a validated judge, and a controlled comparison. The initial targets are answer correctness and explanation sufficiency; causal faithfulness of hidden computation lies outside the claim.

2 Related Work

Concise reasoning and budget control. Sketch-of-Thought uses constrained reasoning paradigms to shorten outputs [4]; s1 controls test-time reasoning through budget forcing [5]. These motivate explicit budget sweeps, but do not supply the premise-retention criterion proposed here. SketchThinker-R1 combines cold-start rewriting, style-judge training, and reinforcement learning [3]. Its default reward is $0.5R_{\text{accuracy}} + 0.4R_{\text{format}} + 0.1R_{\text{style}}$, rather than style alone. A fixed coefficient does not require equal trace lengths, and the paper also examines reward schedules. GPT-5 rewriting could contribute to its gains, but this is a hypothesis requiring controls, not an established cause.

Verification and visual dependence. Process supervision evaluates intermediate reasoning rather than only final answers [6]. Evidence that explanations can misrepresent a model's reasons motivates separating explanatory support from causal faithfulness [7]. MathVerse varies the distribution of information between question text and diagrams [8]; MathVista and MathVision assess visual mathematical reasoning [9, 10]. Our

proposed extension is to evaluate explicit visual premises during compression, with independently checked labels and controlled edits. Qwen2.5-VL supplies the multimodal backbone [11], and Group Relative Policy Optimization (GRPO), introduced in DeepSeekMath [12], supplies the policy-training procedure.

3 Proposed Methodology

Task and operational criterion. Given image I and question q, the answering policy π_θ produces trace z and answer a. Let $L(z)$ count reasoning tokens and B denote a target budget. For each annotated problem, reviewers construct an admissible derivation graph G whose nodes are visual premises, textual premises, and deductions. Visual nodes carry region references or measurable relations. A premise is necessary relative to this derivation if removing it breaks the derivation's support for the answer. Reviewers will accept valid alternative derivations rather than require a single reference wording.

For a candidate trace, $P_V(z)$ measures the fraction of required visual premises stated explicitly and grounded correctly. Similarly, $P_N(z)$ covers necessary nonvisual steps. Let $D(z)$ indicate a valid derivation using only stated premises, question facts, and declared elementary rules. Define

$$S(z;I,q,G) = \mathbf{1}[P_V(z) = 1]\,\mathbf{1}[P_N(z) = 1]\,\mathbf{1}[D(z) = 1]. \tag{1}$$

The image verifies stated visual facts; it must not silently supply missing premises. A correct final answer alone cannot satisfy Eq. 1. For example, "$x = 180 - 40 - 60 = 80$" lacks support if the two angle readings occur only in the image and are never stated. Redundant visual statements may be deleted when a complete admissible derivation remains.

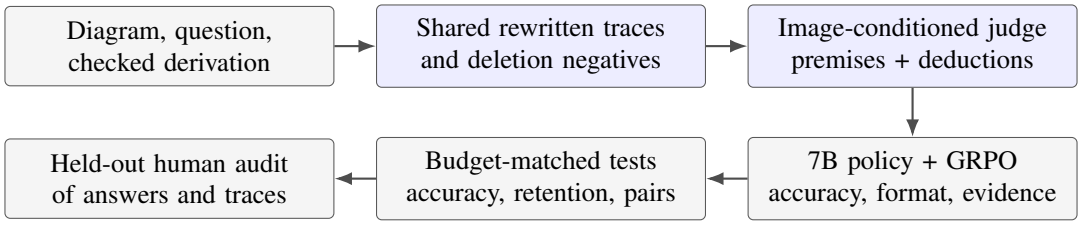


Figure 1: Proposed pipeline. Training annotations supervise the judge; separate development annotations calibrate it. Held-out human evaluation checks both the judge and generated explanations without feeding test labels into training.

Judge architecture and training. A separate Qwen2.5-VL-3B judge will receive I, q, z and predict structured premise-grounding and deduction-validity labels, followed by a sufficiency probability $s_\phi(I,q,z)$. Reference graphs create training labels but are unavailable to the judge at inference. Supervised token cross-entropy will train its structured outputs. Training examples will include supported sketches, single-premise deletions, incorrect visual readings, invalid deductions, and redundant-step deletions. Length- and format-matched positives and negatives will discourage superficial shortcuts. Correct-answer traces with missing evidence are essential negative examples.

Two annotators will independently label the human-audited subset, with adjudication and agreement reported separately for grounding and deduction validity. Development data will calibrate probabilities and choose a threshold τ targeting at most 10% false acceptance and at least 70% recall of sufficient traces. Held-out precision, recall, calibration error, and false acceptance will test whether that target transfers. Missing either development target will trigger additional labeling before policy training.

Compression and policy optimization. All principal reward arms will start from the same Qwen2.5-VL-7B-Instruct checkpoint, fine-tuned on identical compressed traces from one frozen rewriter. Candidate deletions provide judge training data; the answering policy subsequently learns to generate sketches directly.

The proposed reward is

$$R = 0.5\mathbf{1}[a = a^*] + 0.4F(z,a) + 0.1E_\phi(I,q,z) - \lambda \max(0, L(z)/B - 1), \tag{2}$$

where a^* is the checked answer, F checks response format, and $E_\phi = s_\phi \mathbf{1}[s_\phi \geq \tau]$. GRPO will use groups of four rollouts, within-group standardized rewards, and a KL penalty to the cold-start policy. A style arm replaces E_ϕ with a text-only style score; a zero-component arm sets it to zero. All use the same length penalty and development-only coefficient search. Budgets $B \in \{64, 128, 256, 512\}$ are experimental targets, not guaranteed achieved lengths. The judge is used during training only; deployment requires the answering model alone.

4 Experimental Plan

Data and leakage controls. We plan 2,400 programmatically rendered geometry and bar-chart problems, split into 1,600 training, 300 development, and 500 test problems by template family. Generator programs supply reference facts and solutions. Human review will cover 600 training, 100 development, and 200 test problems, including candidate traces for judge supervision and evaluation. All reported test premise scores will use adjudicated annotations. We will build 200 answer-changing test image pairs and 100 nuisance-edit pairs from test originals, with both members human-checked. Edits will preserve problem validity and all unrelated facts. Related images, templates, and alternative traces will remain in the same split.

Table 1: Planned datasets, evaluation measures, and comparisons. Public benchmarks remain outside training and model selection.

Dataset / role	Metrics	Baselines / comparison
Rendered diagrams; training, development, test	Answer accuracy; visual NPR; nonvisual retention; human sufficiency	Shared-data style reward, evidence reward, zero component
Checked image pairs; test only	Both-correct rate; evidence consistency; nuisance invariance	Same three reward arms; judge without image
MathVista testmini; MathVision test; MathVerse testmini [9, 10, 8]	Official answer scoring; reasoning tokens; end-to-end latency; difficulty groups where available	Long reasoning; direct answer; constrained budget; Sketch-of-Thought prompting [4]

Measurements and falsifiable tests. RQ1 will compare deletion rates $d_V = 1 - P_V$ and $d_N = 1 - P_N$ on annotated original-to-compressed traces. Each denominator is the corresponding number of necessary units in the original derivation, not all deleted tokens. RQ2 will measure answer accuracy, visual necessary-premise retention (NPR), and human-rated sufficiency separately. Paired-image correctness requires both answers to be correct. Evidence consistency requires each regenerated trace to state premises true in its own image and support its own answer. Nuisance edits test answer invariance. Unchanged wording across an answer-changing pair is not itself a failure, since a valid relational instruction can apply to both images.

Fair comparisons and uncertainty. Development operating points will match achieved average reasoning-token counts within 5%. Unmatched points will carry no matched-budget claim. The public-test evaluation will use frozen settings and one answer per item, with parser failures counted as incorrect. Latency will include image prefill and answer decoding on fixed hardware at batch size one, after warm-up. MathVision's supplied difficulty levels and annotation-defined derivation-depth groups will support prespecified stratified analysis. Three policy-training seeds and paired bootstrap 95% confidence intervals will quantify uncertainty, resampling original problem families together with their edits and versions and reporting seed variation. A pilot will estimate the detectable effect; inconclusive intervals will not establish equivalence.

For RQ3, the primary evidence-versus-style contrast holds rewritten examples, cold-start weights, decoding, and hyperparameter-search allowance fixed. Training will be capped by measured compute, includ-

ing rollout generation and judge inference, rather than equal update counts alone. We will report processed text/image tokens, estimated FLOPs, accelerator-hours, and separate preprocessing, cold-start, judge-training, and RL costs. Original traces and same-rewriter length-preserving rewrites will form secondary controls under the same outcome-only objective. Their content changes will be audited: preserving length does not isolate style. These controls probe rewriting effects without claiming that GPT-5 rewriting caused the source paper's gains. Image-ablated judges and removal of the evidence component will test which supervision matters.

5 Expected Results & Timeline

We predict that text-only compression will delete a larger proportion of necessary visual premises than necessary nonvisual steps. We further hypothesize that evidence supervision will improve NPR and accuracy over style supervision at matched budgets, with larger benefits on deeper derivations. These predictions are not findings. No positive accuracy difference, poor held-out judge precision, or gains disappearing under compute controls would weaken the proposed advantage. Improved retention without improved accuracy would support explanation quality alone.

The six-month plan assumes access to four 80GB accelerators with parameter-efficient policy tuning; availability is not yet confirmed. A month-one throughput pilot will set the rollout ceiling. If resources are insufficient, we will reduce the number of budget settings before reducing seeds or removing the principal reward contrast. Judge errors, annotation ambiguity, and synthetic-to-public transfer are the main risks. Blinded human audits, acceptance of alternative proofs, and separate public-benchmark results will bound the conclusions. Planned outputs are code, annotation rubrics, permitted data and edits, judge calibration records, and accuracy–cost curves, including null results.

Table 2: Month-by-month milestones and decision points.

Month	Activities and deliverables
1	Reproduce baselines; pilot compute; freeze splits, scoring, and annotation rubric.
2	Generate and annotate diagrams; adjudicate pairs; pilot deletion-rate analysis.
3	Train and calibrate judges; audit negatives; check acceptance and recall targets.
4	Train principal reward arms with three seeds; freeze budget operating points.
5	Run controls, ablations, and held-out tests; analyze uncertainty and difficulty.
6	Complete human audit, failure analysis, reproducibility package, and report.

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